

CH4203: Organic Synthesis Laboratory

Expt. No. 6: Copper-Catalyzed Click Chemistry of an Enantioenriched Azide with a Terminal Alkyne

Home Assignment

Due date: 7 days after the expt. done

(Q1) Write down the mechanism of Cu(I)-catalyzed Click reaction of benzyl azide with phenylacetylene. Why a strong base is not required for this transformation?

[1+2+2 = 5]

(Q2) The Click Chemistry can be done using Cu(II)SO₄·5H₂O in the presence of Sodium Ascorbate (Sodium salt of Ascorbic Acid). How Cu(II)-Species is converted to Cu(I) in the reaction condition?

[1+2+2 = 5]